

STUDY GUIDE

Technical Research: Deepening Handbook

A complete, concept-first reference — prepared for Saikumar Kaleru.

certifications / study / Finance • 2026

Prepared for Saikumar Kaleru. Read top to bottom; each part builds on the last. Bold terms are the vocabulary interviewers listen for.

PART 1 — FOUNDATIONS OF TECHNICAL ANALYSIS

1.1 What technical analysis really is

Technical analysis (TA) is the study of **price and volume data** — usually on charts — to forecast the *future direction* of a market. It does not care *what* a company does or *whether* it is "cheap." It only studies **how the price has behaved** and what that implies next.

A technical analyst believes the chart already contains the collective decisions of every buyer and seller — institutions, retail, algos — and that those decisions leave **repeatable footprints** because human emotion (fear and greed) repeats.

1.2 The three core assumptions (the philosophy)

1. **The market discounts everything.** Every known fact — earnings, news, interest rates, sentiment — is already reflected in the price. So studying price is studying all of it at once.
2. **Prices move in trends.** Once a direction is established, it is more likely to continue than reverse. This is *the* reason TA works — it lets you ride trends.
3. **History repeats itself.** Chart patterns and reactions recur because mass psychology is constant across decades.

1.3 Technical vs Fundamental vs Quantitative

- **Fundamental analysis** answers **WHAT to buy** — it values a business (earnings, growth, debt, management, macro). Time horizon: months to years.
- **Technical analysis** answers **WHEN to buy or sell** — timing entries and exits from price behaviour. Time horizon: minutes to months.
- **Quantitative analysis** uses **statistics and code** to find and test edges systematically (your backtester sits here).
- A strong analyst blends them: *fundamentals pick the stock, technicals time the trade, quant validates the rule.*

1.4 Dow Theory — the foundation everything is built on

Charles Dow's six tenets are the bedrock of TA: 1. **The averages discount everything** (same as assumption 1). 2. **The market has three trends:** *Primary* (months–years, the tide), *Secondary* (weeks–months, corrective waves), *Minor* (days, ripples). 3. **Primary trends have three phases:** **Accumulation** (smart money quietly buys), **Public Participation** (trend-followers pile in, biggest move), **Distribution** (smart money sells to the late crowd). 4. **The averages must confirm each other** — e.g., an index and a related index should move together to confirm a trend. 5. **Volume must confirm the trend** — volume should expand in the direction of the primary trend. 6. **A trend stays in force until a clear reversal** — don't call a top/bottom prematurely.

1.5 Types of charts

- **Line chart** — connects closing prices. Clean, good for seeing the big trend.
- **Bar chart (OHLC)** — each bar shows Open, High, Low, Close.
- **Candlestick chart** — same OHLC data but visual and colour-coded; the global standard (covered in depth in Part 2).

- **Heikin-Ashi** — averaged candles that smooth noise and make trends visually obvious.
- **Renko / Point & Figure** — plot price *movement* (not time); filter out noise to highlight S/R and trends.

1.6 Timeframes and "multiple timeframe analysis"

The same chart looks different on different timeframes. A pro **always checks more than one**: - **Higher timeframe (e.g., weekly/daily)** = the *context / dominant trend*. - **Lower timeframe (e.g., hourly/15-min)** = the *entry timing*. - **Rule**: trade in the direction of the higher timeframe trend, time the entry on the lower one. Conflicting timeframes = lower-conviction trade.

PART 2 — PRICE ACTION & CHART READING

2.1 Candlesticks — full anatomy

Each candle covers one period and shows four prices: - **Open** and **Close** form the **real body**. - The thin lines above/below are **wicks** / **shadows**, marking the **High** and **Low**. - **Bullish candle** (green/white): close > open. **Bearish candle** (red/black): close < open. - **A long body** = strong conviction; **a small body** = indecision; **long wicks** = rejection of a price level.

Single-candle patterns

- **Doji** — open ≈ close (tiny body). Indecision; potential reversal, especially after a long trend.
- **Hammer** — small body at the top, long lower wick ($\geq 2 \times$ body). After a downtrend = **bullish reversal** (buyers rejected lower prices).
- **Hanging Man** — same shape as hammer but after an *uptrend* = bearish warning.
- **Shooting Star** — small body at bottom, long upper wick, after an *uptrend* = **bearish reversal**.
- **Marubozu** — full body, no wicks = extreme conviction in that direction.

Two/three-candle patterns

- **Bullish Engulfing** — a big green candle completely engulfs the prior red body = strong bullish reversal.
- **Bearish Engulfing** — mirror image = bearish reversal.
- **Piercing Line** / **Dark Cloud Cover** — partial engulfing reversals.
- **Morning Star** (bullish) / **Evening Star** (bearish) — three-candle reversals: big trend candle, small indecision candle, big opposite candle.
- **Three White Soldiers** / **Three Black Crows** — three strong candles in a row confirming a reversal.

Interpretation rule: candlestick signals matter most **at a level** (support/resistance) and **with volume** — a hammer at support on high volume is far stronger than one in mid-air.

2.2 Support and Resistance — the most important concept in TA

- **Support** = a price zone where **demand** repeatedly overwhelms supply and stops a fall. A "floor."
- **Resistance** = a price zone where **supply** overwhelms demand and stops a rise. A "ceiling."
- **Why they exist:** memory. Traders remember prices where they bought/sold and act again there.
- **Role reversal (polarity):** once **broken**, old resistance becomes new support, and old support becomes new resistance. This is one of the most reliable behaviours in markets.
- **Types of S/R:** horizontal levels (swing highs/lows), trendlines, moving averages, Fibonacci levels, round numbers (e.g., Nifty 24,000), and pivot points.
- **Strength factors:** the more times a level is tested, the more significant it is — *but* every test drains it, and eventually it breaks.

2.3 Trendlines and channels

- **Uptrend line** — drawn under **higher lows**; acts as support.
- **Downtrend line** — drawn over **lower highs**; acts as resistance.
- **Channel** — two parallel trendlines containing price; buy near the lower rail, sell near the upper rail in a sideways/rising channel.

- A trendline needs **at least two touches** to draw and a **third** to confirm. A decisive break warns of a trend change.

2.4 Chart patterns

Chart patterns are recognisable shapes that tend to resolve in a predictable direction. They fall into **reversal** and **continuation** groups.

Reversal patterns (trend likely to flip)

- **Head & Shoulders** — three peaks, the middle (head) highest, with a "neckline." Break below the neckline = bearish target $\approx$ head-to-neckline distance projected down. **Inverse H&S** = bullish bottom.
- **Double Top (M) / Double Bottom (W)** — two failed attempts at a level = reversal.
- **Triple Top / Bottom** — stronger version.
- **Rounding Top / Bottom (saucer)** — slow, gradual reversal.

Continuation patterns (trend likely to resume)

- **Triangles:** *Ascending* (flat top, rising lows — bullish bias), *Descending* (flat bottom, falling highs — bearish bias), *Symmetrical* (converging — breakout in trend direction).
- **Flags & Pennants** — short pause after a sharp move ("flagpole"), then continuation. Target $\approx$ flagpole height.
- **Wedges** — rising wedge = bearish, falling wedge = bullish (counter-intuitive — learn this).
- **Rectangles** — sideways consolidation between S/R, then breakout.
- **Cup & Handle** — a rounded base then a small pullback (handle), then breakout = bullish.

Gaps

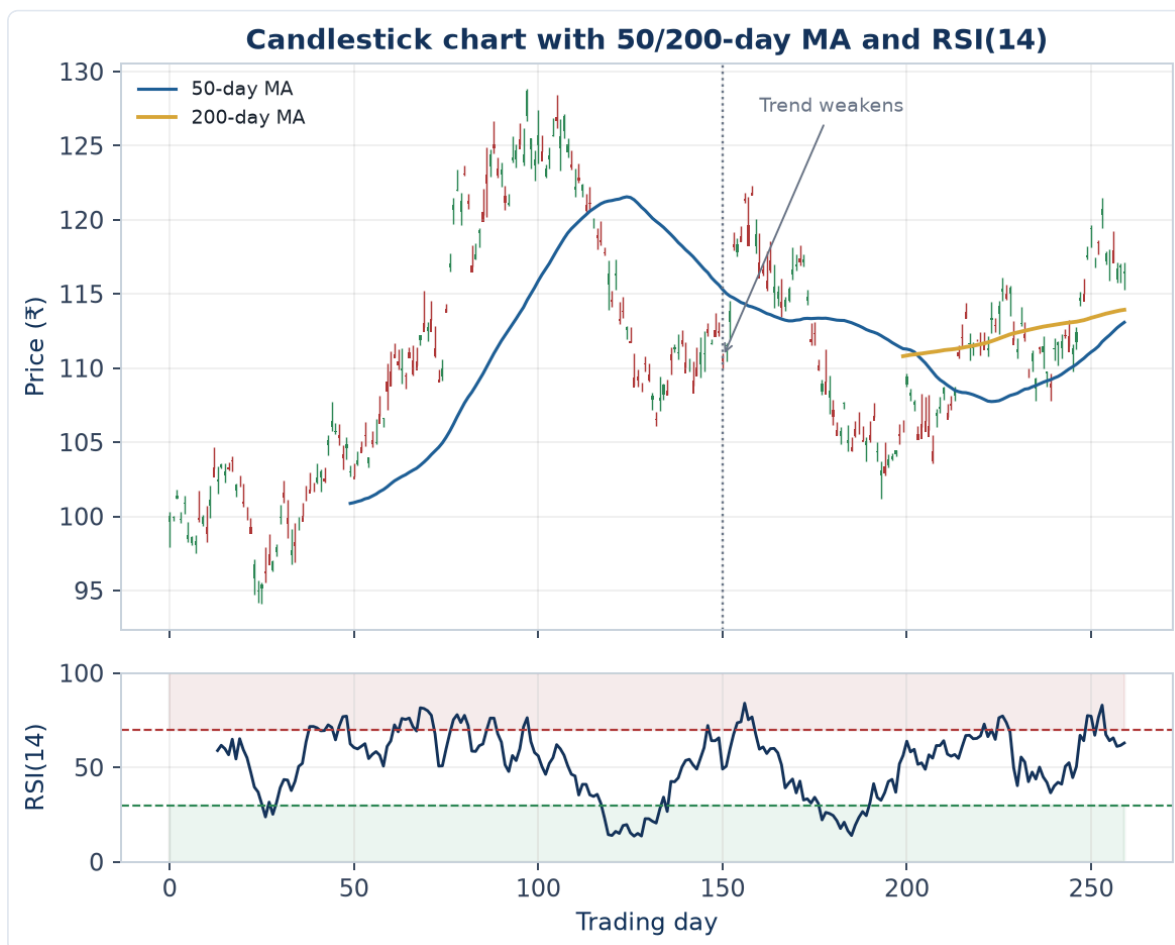
- A **gap** is empty space where price jumps between sessions. Types: **breakaway** (starts a move), **runaway/continuation** (mid-trend), **exhaustion** (end of move), **common** (noise). Gaps often "fill" later.
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PART 3 — TECHNICAL INDICATORS (IN DEPTH)

Indicators are math applied to price/volume to make a signal clearer. They fall into four families: **trend**, **momentum**, **volatility**, **volume**. *Use them to confirm price action, never blindly.*

3.1 Trend indicators

- **Moving Average (MA):** average of the last N closes; smooths noise.
- **SMA** weights all days equally; **EMA** weights recent days more (faster, less lag).
- Use: price above MA = uptrend; MA slope = trend direction; MAs act as dynamic S/R.
- **Crossovers:** short MA crossing above long MA = bullish. **Golden Cross** = 50-DMA above 200-DMA (major bullish). **Death Cross** = opposite.
- **MACD (Moving Average Convergence Divergence):** $(12\text{-EMA} - 26\text{-EMA}) = \text{MACD line}$; its 9-EMA = signal line; the gap = histogram.
- Bullish when MACD crosses above signal; histogram shows momentum strength; **MACD divergence** warns of reversals.
- **ADX / DMI (Average Directional Index):** measures **trend strength** (0–100), not direction. **ADX > 25 = strong trend**; **< 20 = weak/range**. +DI and –DI lines show direction. *Crucial because indicators like RSI work differently in trends vs ranges.*
- **Parabolic SAR:** dots that flip above/below price; trailing-stop and trend-direction tool.
- **Supertrend:** ATR-based trend line, very popular in Indian markets for intraday F&O.



Reading this chart top to bottom: the 50-day MA (blue) tracks the shorter-term trend and reacts faster to price than the flatter 200-day MA (gold) once it appears after enough history accumulates — a golden/death cross is exactly the 50-day line crossing the 200-day line. Around the marked point the uptrend loses momentum: price stops making clean higher highs, the 50-day MA flattens and rolls over. The RSI panel below shows the same story from a momentum angle — RSI oscillates into the overbought zone (>70, red shading) during the strongest leg of the uptrend and into oversold (<30, green shading) during the subsequent pullback, exactly the confluence (trend + moving average + momentum) Part 3.7's "golden rule of indicators" asks for. - **Ichimoku Cloud**: an all-in-one system (trend, S/R, momentum) using a "cloud" (Kumo); price above the cloud = bullish.

3.2 Momentum indicators (oscillators)

Oscillators move within a range and flag **overbought/oversold** and **divergence**. Best in **sideways** markets. - **RSI (Relative Strength Index)**: 0–100 (default 14). >70 overbought, <30 oversold. In strong uptrends RSI can *stay* overbought — don't short blindly. **Divergence** (price up, RSI down) is its most powerful signal. The 50 line = trend bias. - **Stochastic Oscillator**: compares close to the high-low range (%K and %D lines), 0–100, >80/<20 zones. Good for ranges and divergence. - **CCI (Commodity Channel Index)**: measures deviation from average price; ± 100 thresholds. - **Williams %R**: like stochastic, inverted scale (0 to –100). - **ROC / Momentum**: raw rate of price change. - **MFI (Money Flow Index)**: "volume-weighted RSI" — momentum that includes volume.

3.3 Volatility indicators

- **Bollinger Bands**: 20-SMA ± 2 standard deviations. Bands **widen** in volatility, **narrow** ("squeeze") before big moves. Price riding the upper band = strength; tags of the lower band in a range = bounce candidates. **Bollinger squeeze** is a classic breakout setup.
- **ATR (Average True Range)**: the average daily range (true range accounts for gaps). Pure **volatility gauge** — does not give direction. Used to **size stop-losses** (e.g., 1.5–2 $\times$ ATR) and position sizing so each trade risks a similar amount.
- **Keltner Channels**: like Bollinger but ATR-based; squeeze strategies combine both.
- **Donchian Channels**: highest high / lowest low over N periods — the basis of classic breakout systems.

3.4 Volume indicators

Volume is the fuel; price is the car. Moves on rising volume are trustworthy. - **OBV (On-Balance Volume)**: cumulative volume (added on up days, subtracted on down days); confirms trends and shows accumulation/distribution via divergence. - **VWAP (Volume-Weighted Average Price)**: the average price weighted by volume during the day — institutions benchmark fills against it; a key **intraday** S/R level. - **Volume Profile**: shows volume **by price level** (not time), revealing high-volume nodes (strong S/R) and the **Point of Control (POC)**. - **A/D Line (Accumulation/Distribution)** and **Chaikin Money Flow (CMF)**: measure buying vs selling pressure.

3.5 Fibonacci tools

Based on the ratios 0.236, 0.382, 0.5, 0.618 (golden ratio), 0.786. - **Fibonacci Retracement**: after a move, pullbacks often reverse at these % levels. **38.2%–61.8%** is the key zone; 61.8% is the most watched. - **Fibonacci Extension**: projects targets beyond the prior move (1.272, 1.618) — used to set profit targets. - **Confluence**: Fib levels lining up with S/R or a moving average = high-probability zone.

3.6 Pivot points

Calculated from the prior day's High, Low, Close: a central **Pivot (P)** plus support (S1, S2, S3) and resistance (R1, R2, R3) levels. Hugely popular **intraday** for indices and F&O; price above P = bullish bias for the day.

3.7 The golden rule of indicators

Indicators lag — price leads. Use 2–3 *non-redundant* indicators (e.g., one trend + one momentum + volume), look for **confluence**, and let **price action at key levels** make the final call. Stacking five momentum indicators that all say the same thing is false confidence.

PART 4 — MARKET THEORIES & FRAMEWORKS

4.1 Divergence (master this)

Divergence = price and an oscillator (RSI/MACD) disagree. - **Regular bearish divergence**: price makes a **higher high**, indicator makes a **lower high** → uptrend weakening, possible top. - **Regular bullish divergence**: price makes a **lower low**, indicator makes a **higher low** → downtrend weakening, possible bottom. - **Hidden divergence**: signals trend *continuation* (e.g., price higher low, indicator lower low in an uptrend = trend resumes).

4.2 Elliott Wave Theory

Markets move in repetitive **wave cycles** driven by crowd psychology: - A trend unfolds in **5 waves** (1-2-3-4-5, "impulse") followed by a **3-wave correction** (A-B-C). - Waves are **fractal** (the same pattern appears on every timeframe). - Powerful but subjective; used to anticipate where a move is in its life-cycle.

4.3 The Wyckoff Method

Explains how "smart money" operates through cycles: **Accumulation** → **Markup** → **Distribution** → **Markdown**. Teaches you to read **accumulation/distribution ranges**, springs (false breakdowns), and the relationship between price and volume to follow institutional footprints.

4.4 Market cycles & sector rotation

- Markets cycle with the economy: early-cycle, mid, late, recession. Different **sectors** lead at each stage (e.g., financials early, energy/materials late, defensives in downturns).
- **Relative strength analysis** (comparing a stock/sector to the index) tells you what's leading — analysts rotate coverage toward leaders.

4.5 Sentiment & breadth

- **India VIX** — the "fear gauge"; high VIX = fear/volatility, often near bottoms.
 - **Put-Call Ratio (PCR)** — options sentiment (covered in Part 5).
 - **Market breadth** — **Advance/Decline line**, % of stocks above their 200-DMA, new highs vs new lows. Breadth confirms or warns against an index move (a rally on narrow breadth is fragile).
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PART 5 — DERIVATIVES: FUTURES & OPTIONS (F&O)

Your D.E. Shaw desk was F&O — expect deep questions here.

5.1 What derivatives are

A **derivative** derives its value from an **underlying** (stock, index, commodity). Used for **hedging**, **speculation**, and **leverage**. Traded in **lots** (fixed quantity), with monthly/weekly **expiries**.

5.2 Futures

- A **futures contract** is a binding agreement to buy/sell the underlying at a set price on a future date.
- **Long futures** = bullish; **short futures** = bearish. Leverage via **margin** (you post a fraction of contract value).
- **Basis** = futures price – spot price. **Contango** = futures > spot (normal); **Backwardation** = futures < spot (often tight supply, common in commodities).
- **Cost of carry** = interest + storage – dividends; explains the futures-spot gap.
- **Rollover** — moving a position to the next expiry; high rollover % signals trend conviction.

5.3 Options — the core

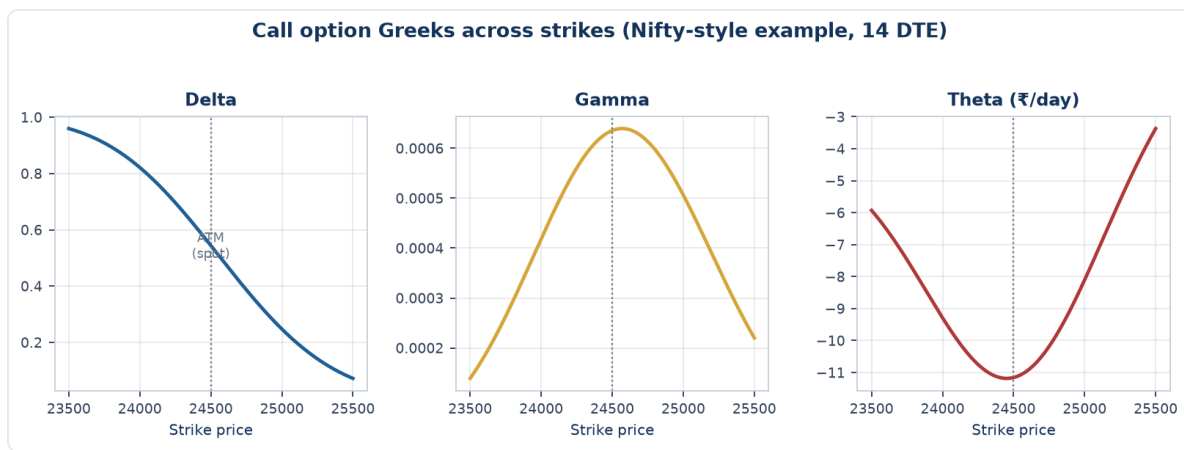
An **option** is the **right, not the obligation**, to buy/sell the underlying at a **strike price** before/at **expiry**, for a **premium**. - **Call option** = right to **buy** → bought when **bullish**. - **Put option** = right to **sell** → bought when **bearish**. - **Buyer** pays premium, has limited risk (premium) and unlimited upside. **Seller (writer)** receives premium, has limited profit but large risk.

Moneyiness

- **ITM (In-the-Money)**: has intrinsic value (call strike < spot; put strike > spot).
- **ATM (At-the-Money)**: strike ≈ spot.
- **OTM (Out-of-the-Money)**: no intrinsic value, only time value.
- **Premium = Intrinsic value + Time value**. Time value decays to zero by expiry.

5.4 The Option Greeks (know all five)

- **Delta** — change in option price per ₹1 move in the underlying (also ≈ probability of expiring ITM). Calls 0→1, puts 0→-1.
- **Gamma** — the rate of change of delta; highest for ATM options near expiry (the "acceleration").
- **Theta** — **time decay**: how much value the option loses each day. Negative for buyers, positive for sellers.
- **Vega** — sensitivity to a 1% change in **implied volatility**. Long options are long vega.
- **Rho** — sensitivity to interest rates (least important intraday).



Read left to right: **Delta** falls smoothly from near 1 (deep ITM, behaves like the underlying) to near 0 (deep OTM, unlikely to ever pay off) as strike rises past spot. **Gamma** peaks exactly at-the-money — the point where a small underlying move can flip the option between likely-worthless and likely-valuable, so Delta itself is most sensitive there. **Theta** is most negative at-the-money too — the option with the most uncertainty (closest to a coin-flip outcome) has the most time value to lose each day, which is exactly why ATM options decay fastest as expiry approaches.

5.5 Implied Volatility (IV) — you worked on IV surfaces

- **IV** is the market's **expectation of future volatility**, backed out of option prices (via Black-Scholes). High IV → expensive options.
- **IV vs HV:** implied (forward-looking) vs historical/realised (backward-looking).
- **IV Rank / IV Percentile** — where current IV sits versus its own past range; tells you if options are "cheap" or "expensive."
- **Volatility Skew / Smile** — IV differs across strikes; equity index puts usually have higher IV (crash protection demand). Plotting IV across strikes **and** expiries gives the **volatility surface** (your project).
- **Trading logic:** buy options when IV is low, sell premium when IV is high.

5.6 Open Interest (OI) & option-chain analysis

- **Open Interest** = total outstanding contracts. **Rising OI + rising price** = fresh longs (strong); **rising OI + falling price** = fresh shorts; **falling OI** = unwinding.
- **Option chain** = the full grid of calls/puts by strike with price, OI, IV, volume.
- **Put-Call Ratio (PCR)** = put OI ÷ call OI. **PCR > 1** = bearish positioning (often contrarian bullish); **< 0.7** = bullish/greedy.
- **Support/Resistance from OI:** the strike with the **highest call OI** acts as resistance; **highest put OI** acts as support.
- **Max Pain** = the strike where option buyers lose the most / writers gain the most; price often gravitates there near expiry.

5.7 Core option strategies

- **Covered Call** — hold stock + sell a call: income, caps upside.
- **Protective Put** — hold stock + buy a put: insurance.
- **Long Straddle** — buy call + put at the **same strike**: profit from a **big move either way** (event/earnings plays); needs a move bigger than the combined premium.

- **Long Strangle** — same idea with **different OTM strikes**: cheaper, needs a bigger move.
- **Bull Call Spread / Bear Put Spread** — buy one option, sell another to **reduce cost** and cap risk/reward.
- **Iron Condor** — sell an OTM call spread + put spread: profit when price stays **range-bound** (income in low volatility).
- **Butterfly** — bet on price pinning a specific strike.

5.8 Worked example — reading the Greeks on a real position

Nifty spot at 24,500. You buy 1 lot (75 units) of a 24,600 Call, premium ₹180, with Delta 0.42, Gamma 0.003, Theta -8.5, Vega 12.*

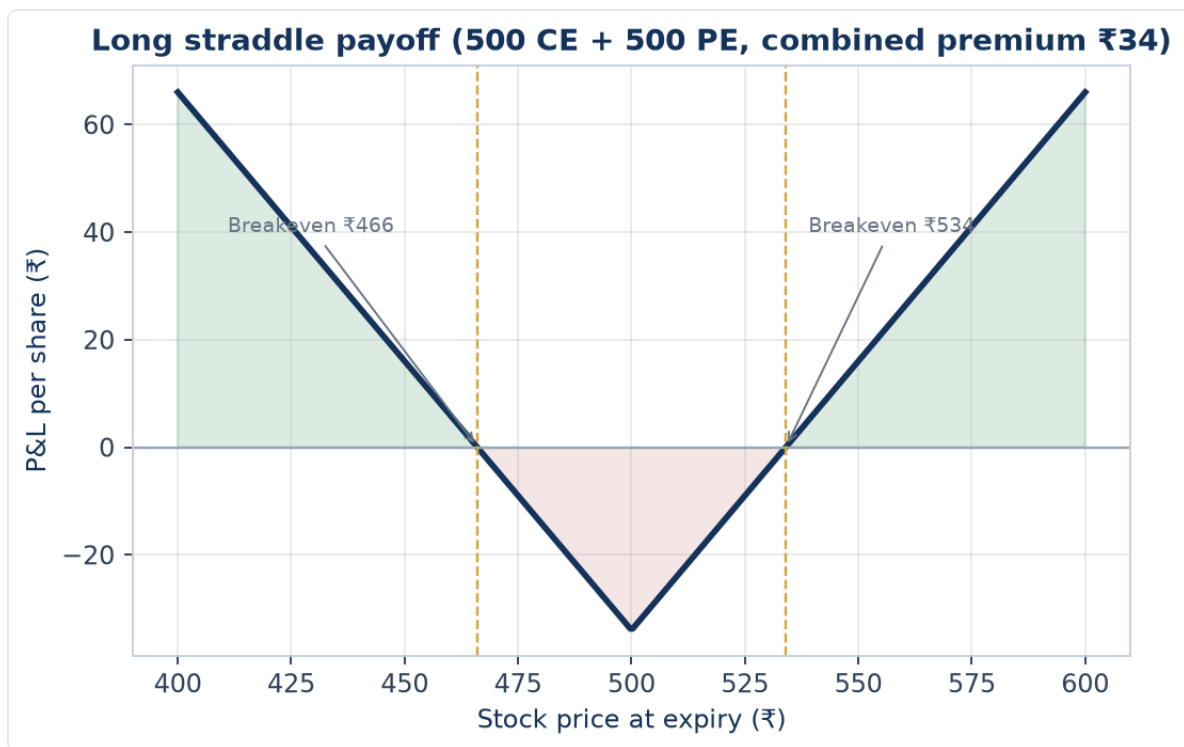
Interpreting each Greek on this position: - **Delta 0.42** — if Nifty rises 100 points, this option gains $\approx 0.42 \times 100 = ₹42$ per unit, or $₹42 \times 75 = ₹3,150$ on the lot (before Gamma's second-order effect). Delta also approximates a 42% probability of expiring ITM. - **Gamma 0.003** — as Nifty moves, Delta itself changes by ≈ 0.003 per point. After a 100-point rise, new Delta $\approx 0.42 + (0.003 \times 100) = 0.72$ — the position gains sensitivity *faster* as it moves further ITM, which is why Gamma is called the "acceleration" of the option's price move. - **Theta -8.5** — this position loses $\approx ₹8.5$ per unit *per day* purely from time decay if nothing else changes, i.e. $₹8.5 \times 75 = ₹637.50/\text{day}$ — a buyer is fighting this decay every single day the trade is held, which is why option buying requires the underlying to move meaningfully and soon, not just eventually. - **Vega 12** — a 1-percentage-point rise in implied volatility adds $\approx ₹12$ per unit (₹900 on the lot); a fall in IV costs the position the same amount even if the underlying doesn't move at all — this is why buying options right before a scheduled event (results, a central-bank decision) is risky even if the *direction* call is correct: IV often collapses ("IV crush") right after the event, working against a long-option holder even on a correct directional move.

The synthesis a TRA must be able to state out loud: "This is a long-delta, long-gamma, negative-theta, positive-vega position — it wants the underlying to move up, fast, with rising or at least stable implied volatility; every day that passes without a move, and any drop in IV, works against it."

5.9 Worked example — a long straddle P&L around an earnings event

Stock at ₹500 ahead of results. Buy the 500 Call for ₹18 and the 500 Put for ₹16 — total premium paid ₹34. Breakevens: $500 + 34 = ₹534$ (upside) and $500 - 34 = ₹466$ (downside).

Model answer. The position profits if the stock moves *outside* the ₹466-₹534 range by expiry (or sooner, from an IV spike), regardless of direction — this is the textbook "big move either way" trade for a binary event like earnings. If the stock closes at ₹560 post-results: the call is worth $560 - 500 = ₹60$, the put expires worthless, net P&L = $60 - 34 = +₹26$ **per share**. If the stock closes at ₹495 (a modest, unremarkable move): the call expires worthless, the put is worth $500 - 495 = ₹5$, net P&L = $5 - 34 = -₹29$ **per share** — a straddle loses money on a move that's "real" in direction but not big enough to clear the combined premium cost, which is precisely why sizing the expected move (often estimated from the option chain's implied volatility itself) against the premium paid is the core skill in event-driven options trading, not just having a directional view.



PART 6 — COMMODITIES

The JD's first line is "Analyze Commodity, Futures & Options."

6.1 The Indian commodity market

- Traded mainly on the **MCX (Multi Commodity Exchange)**; agri commodities on **NCDEX**.
- Categories: **Bullion** (gold, silver), **Energy** (crude oil, natural gas), **Base metals** (copper, zinc, aluminium, lead), **Agri** (cotton, soybean, etc.).
- Indian prices **track global benchmarks** and adjust for the **USD/INR** rate.

6.2 Gold

- **Drivers:** the **US dollar** (inverse — strong USD weakens gold), **real interest rates** (rising real yields hurt gold, which pays no income), **inflation** (hedge), **safe-haven demand** in crises, and central-bank buying.
- Benchmark: **COMEX** (USD/oz); Indian MCX gold also moves with **USD/INR** and import duty.

6.3 Silver

- Part precious metal, part **industrial** (solar, electronics) → more volatile than gold. The **gold-silver ratio** is watched for relative value.

6.4 Crude Oil

- Benchmarks: **WTI** (US) and **Brent** (global). **Drivers:** **OPEC+** supply decisions, geopolitics (Middle East), global demand/growth, US inventory data (EIA), and the USD.
- **Contango/backwardation** in the futures curve signals supply tightness.

6.5 Natural gas & base metals

- **Natural gas** — extremely volatile, weather-driven (heating/cooling demand).
- **Copper ("Dr. Copper")** — a barometer of global economic health due to its industrial use.

6.6 Cross-asset & seasonality

- Commodities have **seasonal** tendencies (e.g., gold demand around Indian festivals/weddings; gas in winter).
 - Same TA toolkit (trend, S/R, RSI, MACD, MAs, Fibonacci) applies directly to commodity charts.
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PART 7 — INDICES & THE BROADER MARKET

7.1 Indian indices

- **Nifty 50** — 50 largest, most liquid NSE stocks; the national benchmark.
- **Sensex** — BSE's 30-stock benchmark.
- **Bank Nifty** — 12 major banks; **more volatile**, the most traded F&O index in India.
- **Fin Nifty, Nifty Midcap, Nifty Next 50**, and **sectoral indices** (IT, Pharma, Auto, FMCG, Metal) — used for sector rotation and relative strength.

7.2 Global cues

- **GIFT Nifty (formerly SGX Nifty)** — signals the likely opening of Nifty based on overnight global moves.
- **US indices** — **Dow, S&P 500, Nasdaq**; **Asian** (Nikkei, Hang Seng) and **European** opens set the tone.
- **Dollar Index (DXY), US 10-year yield, crude, and FII/DII flows** all drive Indian markets.

7.3 Volatility & breadth

- **India VIX** — expected 30-day volatility; spikes in fear.
 - **Advance/Decline, % above 200-DMA, new highs/lows** — breadth gauges that confirm index health.
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PART 8 — RISK & TRADE MANAGEMENT

This is what separates a professional from a gambler. Interviewers test it hard.

8.1 The stop-loss

A **stop-loss** is a pre-decided exit price that caps your loss if you're wrong. **Non-negotiable.** Types: - **Fixed % stop**, **level-based stop** (below support), **ATR/volatility stop**, **moving-average stop**, **time stop**. - **Trailing stop** — moves with price to lock in profit while letting winners run.

8.2 Risk:Reward and R-multiples

- **Risk:Reward (R:R)** = potential reward ÷ potential risk. Demand $\geq 1:2$ so you can be right less than half the time and still profit.
- **R-multiple** — express every result in units of initial risk (R). A trade that makes 3× your risk = +3R. Thinking in R keeps you objective.

8.3 Position sizing

- **Risk a fixed small % of capital per trade** (commonly 1–2%). Position size = (capital × risk%) ÷ (entry – stop). This ensures no single trade can hurt you and that volatile instruments get smaller size (via wider stops).

8.4 Portfolio-level risk

- **Diversification & correlation** — don't hold five positions that are really the same bet (e.g., all bank stocks + Bank Nifty).
- **Max drawdown discipline, exposure limits, and avoiding over-leverage** in F&O (leverage cuts both ways).

8.5 The money-management rules pros live by

Cut losses quickly, let winners run, never add to a loser hoping it recovers, size to volatility, and **survive first — returns come second.**

8.6 Worked example — position sizing and a full R-multiple trade

Capital: ₹10,00,000. Risk per trade rule: 1.5% of capital. Stock trades at ₹842; support/stop level at ₹810; target at ₹930.

Position sizing:

Risk amount = $10,00,000 \times 1.5\% = ₹15,000$

Risk per share = Entry – Stop = $842 - 810 = ₹32$

Position size = Risk amount / Risk per share = $15,000 / 32 \approx 468$ shares

Round to 468 shares (or down to 460 for a round lot, whichever the broker/market convention requires) — this is the *maximum* size that keeps the pre-defined loss at exactly ₹15,000 (1.5% of capital) if the stop is hit, regardless of how "confident" the analyst feels about the trade; conviction should show up in written rationale, never in oversizing beyond the risk rule.

Risk:Reward and R-multiple:

Reward per share = Target - Entry = 930 - 842 = ₹88
R:R = Reward / Risk = 88 / 32 = 2.75 (i.e. "1:2.75")

This clears the $\geq 1:2$ minimum bar comfortably — meaning the trade can be *wrong* more than a third of the time (specifically, a bit above a 26.7% win rate, from the breakeven formula in 10.4's expectancy calculation) and still be profitable over a large enough sample, which is the entire point of thinking in R-multiples rather than any single trade's outcome.

If the trade hits target: P&L = $468 \times 88 = ₹41,184$, or **+2.75R** relative to the ₹15,000 initial risk. **If the trade hits the stop instead:** P&L = $-468 \times 32 = -₹14,976 \approx -1R$, exactly as sized — the loss was known and bounded *before* the trade was placed, which is the entire discipline this section is testing.

PART 9 — RESEARCH REPORT WRITING & RECOMMENDATIONS

Your actual daily output as a TRA.

9.1 Report cadence

- **Daily** — pre-market note: overnight global cues, key levels for Nifty/Bank Nifty/commodities, top trade ideas.
- **Weekly** — trend review, sector view, positional calls.
- **Monthly** — bigger-picture market outlook and themes.

9.2 Anatomy of a good call

Every recommendation must state: **Instrument • View (Buy/Sell/Hold) • Entry • Target(s) • Stop-loss • Time horizon • Rationale.** - The **rationale** ties together trend, levels, indicators, and (for F&O) OI/IV. - Always give a **stop** and an honest **risk:reward** — a call without a stop is unprofessional.

9.3 Communicating to clients & RMs

- Be **clear and concise** — RMs relay to clients who aren't technical. Lead with the call, then the reason.
 - Talk in **probabilities, not certainties** ("favourable risk-reward to go long above X," not "this will go up").
 - **Track and own your calls** — record hit/miss; credibility is built on a transparent track record.
-

PART 10 — BACKTESTING & QUANTITATIVE BASICS

Your differentiator — but you must understand it, not just run it.

10.1 What backtesting is

Applying a **rules-based strategy to historical data** to estimate how it would have performed — to validate an edge **before** risking capital.

10.2 Methodology & pitfalls

- **No look-ahead bias** — only use data available at decision time (act on *yesterday's* signal).
- **Survivorship bias** — include delisted names, or results are too rosy.
- **Overfitting (curve-fitting)** — tuning a strategy so perfectly to the past that it fails live. Guard with **out-of-sample testing** and **walk-forward analysis**.
- **Costs** — include brokerage, slippage, and taxes, or live results disappoint.

10.3 Performance metrics (define each)

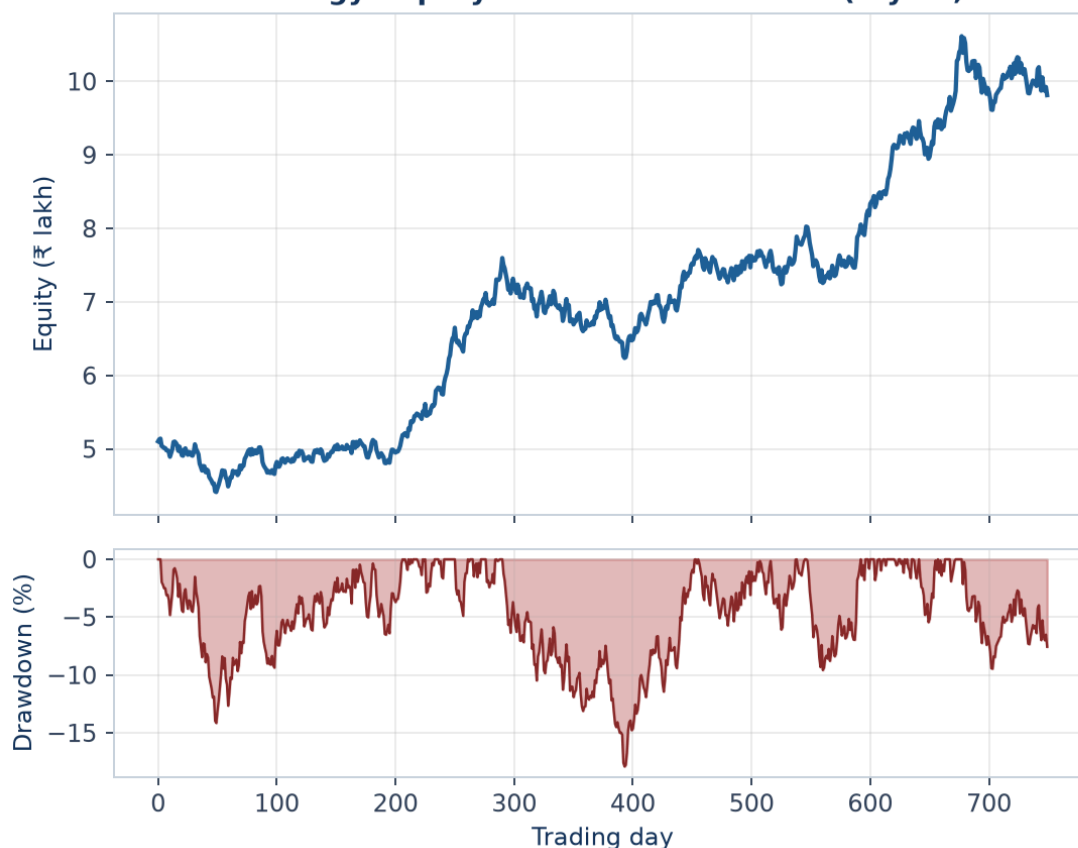
- **CAGR** — smoothed annual growth rate.
- **Sharpe ratio** — excess return ÷ total volatility; risk-adjusted return. >1 good, >2 excellent.
- **Sortino ratio** — like Sharpe but only penalises **downside** volatility (fairer).
- **Maximum Drawdown** — worst peak-to-trough loss; measures pain/risk of ruin.
- **Win rate** — % of profitable trades (high win rate ≠ profitable if losses are huge).
- **Profit factor** — gross profit ÷ gross loss; >1 = profitable.
- **Exposure** — % of time invested.
- **Calmar ratio** — CAGR ÷ max drawdown.

10.4 Statistics you should know

- **Mean, median, standard deviation** (volatility = std of returns), **normal distribution & fat tails** (markets have more extreme moves than "normal" predicts), **correlation** (−1 to +1), **mean reversion vs momentum**, and **probability/expectancy** = (win% × avg win) − (loss% × avg loss).

10.5 Worked example — evaluating a backtested strategy's full metric set

A backtested strategy over 3 years: 240 trades, 96 winners (avg win ₹4,200), 144 losers (avg loss ₹1,800). Starting capital ₹5,00,000, ending capital ₹8,90,000. Worst peak-to-trough equity decline during the period: ₹1,10,000 from a peak of ₹7,20,000. Strategy's daily returns have a standard deviation implying an annualised volatility of 18%; risk-free rate assumed 6%.

Backtested strategy: equity curve and drawdown (3-year, illustrative)**Win rate and expectancy:**

Win rate = $96/240 = 40\%$

Expectancy = $(0.40 \times 4,200) - (0.60 \times 1,800) = 1,680 - 1,080 = +₹600/\text{trade}$ (positive → profitable)

Note the strategy is profitable *despite* a 40% win rate — well below 50% — because winners are more than double the average loser in size, directly illustrating why win rate alone is a misleading metric without pairing it with average win/loss size (10.3's explicit warning).

Profit factor:

Gross profit = $96 \times 4,200 = ₹4,03,200$

Gross loss = $144 \times 1,800 = ₹2,59,200$

Profit factor = $4,03,200 / 2,59,200 \approx 1.56$ (>1 = profitable; a common quality bar is >1.5)

CAGR:

Total return = $8,90,000/5,00,000 = 1.78x$ over 3 years

CAGR = $1.78^{(1/3)} - 1 \approx 21.1\%$

Maximum drawdown:

Max DD = $1,10,000 / 7,20,000 \approx 15.3\%$

A ~21% CAGR against a ~15% max drawdown is a reasonable, though not exceptional, risk-adjusted profile — the next step is checking the **Calmar ratio** ($\text{CAGR} \div \text{Max DD} = 21.1/15.3 \approx 1.38$) to see how the return compares directly to the worst pain endured, which is often more intuitive to a risk-averse allocator than the Sharpe ratio alone.

Sharpe ratio (using the 21.1% CAGR as the return proxy and 18% annualised volatility):

$$\text{Sharpe} \approx (21.1\% - 6\%) / 18\% \approx 0.84$$

Below the ">1 good, >2 excellent" bar from 10.3 — a red flag that, despite a healthy-looking CAGR, this strategy's return isn't especially well-compensated for the volatility it experiences along the way, and an interviewer would expect the candidate to flag this tension (good CAGR, mediocre Sharpe) explicitly rather than only citing the flattering CAGR number.

PART 11 — REGULATION, COMPLIANCE & ETHICS

11.1 NISM Series-XV: Research Analyst

The **SEBI-mandated certification** required to legally publish research/recommendations in India. It covers TA, fundamentals, valuation, and — importantly — **regulations and ethics**.

11.2 SEBI (Research Analysts) Regulations — key duties

- **Disclosure of conflicts** — reveal any holding or interest in a recommended security.
- **No front-running** — never trade ahead of your own published research.
- **Separation** of research from trading/dealing and investment-banking functions.
- **Rational basis & records** — recommendations must be justified and documented; maintain research records.
- **Fair dealing** — no misleading claims, no guaranteed returns, suitability in mind.

11.3 Why it matters in interviews

Showing you understand **disclosures, conflict management, and that research must be compliant** signals maturity beyond just chart-reading.

PART 12 — BEHAVIOURAL FINANCE & MARKET PSYCHOLOGY

Markets are driven by **fear and greed**; TA is, at heart, the study of crowd psychology. - **The emotional cycle**: optimism → euphoria (tops) → anxiety → panic/capitulation (bottoms) → hope → optimism. - **Common biases to know**: **herd mentality**, **FOMO**, **loss aversion** (holding losers, cutting winners early), **confirmation bias** (seeing only what supports your view), **anchoring**, **recency bias**, and **overconfidence**. - **The professional edge** is *discipline* — a written plan, predefined stops, and the emotional control to follow them when the crowd panics.

PART 13 — TOOLS OF THE TRADE

- **Charting: TradingView** (industry standard), broker terminals, MetaTrader, and pro terminals like **Bloomberg / Refinitiv (Reuters)**.
 - **Data & screening:** NSE/MCX option chains, **Chartink / screeners**, Excel for tracking and models.
 - **Coding (your edge): Python** (pandas, yfinance, matplotlib) for backtesting, screening, and automating reports — exactly what your projects demonstrate.
 - **Knowing the workflow:** pre-market prep → global cues + levels → intraday monitoring → end-of-day report → calls tracking.
-

PART 14 — COMMON TRADING SETUPS (putting it together)

- **Trend-following / pullback:** in an uptrend, buy a dip to a moving average or support with the trend.
 - **Breakout:** enter when price clears a key level/range on rising volume; stop just inside the range.
 - **Reversal:** trade a confirmed reversal pattern + divergence at a major level (higher risk).
 - **Mean-reversion:** in a range, fade extremes (buy oversold support, sell overbought resistance).
 - **Momentum:** buy strength/relative-strength leaders; ride until momentum fades.
 - **Event/volatility (F&O):** straddles/strangles around results or events when a big move is expected.
 - **Every setup needs:** a trigger, an entry, a **stop**, a target, and a reason. No exceptions.
-

QUICK REVISION — THE ANALYST'S MENTAL CHECKLIST

1. What's the **trend** on the higher timeframe?
 2. Where are the key **support/resistance** levels?
 3. What do **price action / candles** say at those levels?
 4. Do **indicators** (one trend + one momentum + volume) confirm? Any **divergence**?
 5. For F&O: what do **OI, PCR and IV** say?
 6. Define the trade: **entry, target, stop, risk:reward, horizon.**
 7. **Size** the position to risk a small fixed % .
 8. Write the call **clearly with a rationale** — and **track** it.
-

PART 15 — INTERMARKET ANALYSIS

15.1 What intermarket analysis is and why a TRA needs it

Intermarket analysis studies the relationships *between* asset classes (equities, bonds, commodities, currencies) rather than analysing any single market in isolation — the premise being that these markets are economically linked, and a move in one often leads or confirms a move in another. A technical research analyst who only watches the equity index in isolation misses these cross-asset signals that professional macro/multi-asset desks watch constantly.

15.2 The four core intermarket relationships

- **Bonds and equities:** typically inversely related over the medium term — rising bond yields (falling bond prices) raise the discount rate applied to equity cash flows (the same WACC/CAPM logic from the Valuation chapters) and raise the relative appeal of "safe" fixed income versus equities, pressuring equity valuations, especially high-growth/high-multiple names most sensitive to discount-rate changes.
- **Commodities and currencies:** commodity-exporting economies' currencies (e.g. resource-heavy emerging markets) tend to strengthen when commodity prices rise (higher export revenues) and weaken when they fall — the USD/INR-and-crude-oil relationship is a directly relevant Indian-market example, since India is a large net oil importer: a rising crude price typically pressures the rupee (higher import bill), while a falling crude price tends to support it.
- **US Dollar and commodities:** most globally-traded commodities are priced in USD, so a broadly strengthening dollar mechanically makes commodities more expensive in other currencies, dampening demand and typically pressuring USD-denominated commodity prices lower — an inverse relationship a technical analyst watching gold or crude should track via the Dollar Index (DXY) as a standard cross-check.
- **Bonds and commodities:** rising commodity prices (an inflation signal) often precede or coincide with rising bond yields, as markets price in expected central-bank tightening to control inflation — watching commodity trends can offer an early read on the bond-market direction likely to follow.

15.3 Worked example — using intermarket signals to add conviction to an equity call

A TRA is considering a bullish call on a rate-sensitive Indian real-estate/NBFC stock. Context: US 10-year Treasury yields have been falling for three weeks (bond prices rising), India's 10-year G-Sec yield has followed the same direction, crude oil has fallen 8% over the same period (supportive for the rupee and for India's import-heavy inflation outlook), and the Dollar Index has weakened.

Model answer. All four intermarket signals point the same direction and reinforce the equity thesis: falling global and domestic bond yields directly benefit rate-sensitive sectors (lower discount rates, cheaper financing costs for NBFCs and real-estate developers specifically); falling crude oil supports the rupee and reduces one of the RBI's key inflation-watch inputs, raising the probability of a dovish domestic rate stance that further supports the same rate-sensitive trade; a weaker Dollar Index generally coincides with more supportive conditions for emerging-market risk assets, including Indian equities broadly. **The discipline this example teaches:** a TRA presenting this call to a client or in an interview should explicitly walk through this intermarket confluence, not just point to the stock's own chart — "the equity setup, the falling-yield backdrop, and the currency/commodity picture are all telling the same story" is a materially stronger, more professional case than a chart-pattern observation alone.

15.4 Limits of intermarket analysis

These relationships are historical tendencies, not fixed laws — they can and do break down or reverse for extended periods (a well-known example: the traditional inverse bond-equity relationship has weakened or even reversed during some high-inflation regimes, since both bonds and equities can sell off together when inflation itself is the dominant driver of both markets). A disciplined analyst treats intermarket signals as one input among several (alongside the stock's own technicals and the fundamental backdrop), not a mechanical, always-true rule.

PART 16 — BUILDING A SYSTEMATIC TRADING SYSTEM

16.1 From discretionary rules to a systematic strategy

Everything in Parts 1-14 can be traded **discretionarily** (a human analyst applies judgment to each setup) or **systematically** (a fully coded, rules-based strategy that generates signals without daily human judgment). A systematic approach forces every rule — the trend filter, the entry trigger, the stop, the position size — to be stated with enough precision that a computer can execute it unambiguously, which is itself a valuable discipline even for an analyst who ultimately trades discretionarily, since it exposes vague or inconsistent reasoning that "feels" rigorous in prose but can't actually be coded.

16.2 The core components of a systematic strategy

1. **Universe:** which instruments the strategy trades (e.g. Nifty 50 constituents, or a specific index/commodity).
2. **Signal generation:** the precise, unambiguous rule that triggers a trade (e.g. "50-day EMA crosses above 200-day EMA AND $RSI(14) > 50$ " — note the explicit, codeable thresholds, unlike a discretionary "the trend looks strong").
3. **Entry logic:** exactly when and at what price the trade is taken once a signal fires (next bar's open, at the close of the signal bar, etc. — a detail that matters enormously for backtest realism, per Part 10.2's look-ahead-bias warning).
4. **Exit logic:** both the profit-target and stop-loss rules, and any time-based exit (e.g. close the position if the signal hasn't resolved within N days).
5. **Position sizing rule:** coded exactly as in Part 8.6's worked example, applied systematically rather than judged trade-by-trade.
6. **Portfolio-level rules:** maximum concurrent positions, maximum sector/correlation exposure, and any portfolio-level drawdown circuit-breaker (e.g. "stop taking new signals if the strategy's running drawdown exceeds 15%").

16.3 Worked example — fully specifying a systematic trend-following strategy

Turning the "trend-following / pullback" discretionary setup from Part 14 into codeable rules.

- **Universe:** Nifty 50 constituents, daily bars.
- **Signal:** price above its rising 50-day EMA (defining the uptrend, per Part 3.1) AND $RSI(14)$ between 40-55 (a "healthy pullback, not oversold-breakdown" band, avoiding both an already-extended entry and a genuinely broken trend).
- **Entry:** at the next day's open, following the signal day's close meeting the above condition.
- **Stop:** $1.5 \times$ the 14-day Average True Range (ATR — a volatility-adjusted stop distance, avoiding a fixed % stop that's too tight for a volatile stock and too loose for a calm one) below the entry price.
- **Target:** exit at $3 \times$ the initial ATR-based risk (enforcing a fixed $\geq 1:3$ R:R, per Part 8.2's discipline, systematically rather than judged per-trade).
- **Time exit:** close the position if neither the stop nor target is hit within 20 trading days (preventing capital from sitting indefinitely in a stagnant, directionless trade).
- **Position sizing:** fixed 1% account risk per trade (Part 8.3's formula), capped at a maximum of 8 concurrent open positions to bound total portfolio risk even if many signals fire simultaneously (a

realistic risk during a broad market-wide trend, when many stocks can trigger the same signal at once — an important portfolio-level check a purely per-trade risk rule alone doesn't catch).

16.4 Why explicit specification matters even for a discretionary analyst

Writing out a strategy this precisely (Section 16.3) exposes exactly the kind of vague reasoning that sounds plausible in a research note but wouldn't survive being coded — e.g. "buy on a pullback in an uptrend" leaves the ATR-based stop distance, the exact RSI band, and the position-sizing rule all unstated. A TRA who can move fluidly between the discretionary narrative (Parts 1-9) and the fully-specified systematic version (this Part) demonstrates the quantitative rigor increasingly expected of technical research roles — directly the kind of Python/backtesting differentiator flagged in Part 13's "Tools of the Trade" section.

APPENDIX A — TWO FULLY WORKED TRADE SETUPS, START TO FINISH

A.1 A trend-following pullback setup (cash equity)

Stock XYZ, daily chart: higher highs and higher lows for 4 months (confirmed uptrend). Price pulls back to the rising 50-day EMA at ₹455, which has held as support on the last two pullbacks. RSI on the pullback reads 44 (not oversold, healthy pullback in an uptrend rather than a breakdown). Volume on the pullback is below average (no aggressive selling); a bullish hammer candle forms exactly at the EMA.

Full call, in the "Anatomy of a good call" format from 9.2: - **Instrument:** XYZ equity (cash). - **View:** Buy. - **Rationale:** Established uptrend (Dow Theory higher-highs/higher-lows) intact; pullback to a proven dynamic support (50-day EMA, held twice before) on low, non-aggressive volume; RSI healthy (not overbought going in, not showing bearish divergence); a bullish reversal candle (hammer) confirms buyers stepping in exactly at the level — trend + level + price action + momentum all align (the "confluence" principle from Part 3.7). - **Entry:** ₹458 (on confirmation above the hammer's high). - **Stop:** ₹446 (just below the hammer's low and the EMA — invalidates the setup if hit). - **Target:** ₹512 (prior swing high — the next logical resistance). - **Risk:Reward:** Risk = 458–446 = ₹12; Reward = 512–458 = ₹54; R:R ≈ 1:4.5 — well above the ≥1:2 bar. - **Horizon:** 3-6 weeks (swing trade, matches the daily-chart timeframe used for analysis).

A.2 An event-driven F&O setup (results-day straddle, sized correctly)

A large-cap stock reports quarterly results tomorrow after market close. Current price ₹1,240. The at-the-money straddle (1240 Call + 1240 Put) costs a combined ₹58. Implied volatility has risen sharply into the event (IV Rank in the 85th percentile of its own 1-year range) — a signal options are relatively expensive heading into the event.

Full call: - **Instrument:** 1240 CE + 1240 PE (long straddle), 1-lot. - **View:** Volatility/event play, direction-agnostic. - **Rationale:** A quarterly result is a known, scheduled binary event likely to produce a move larger than the market's average daily range; a long straddle profits from the *size* of the move in either direction. **Caution flagged explicitly:** IV Rank at the 85th percentile means the straddle is pricing in a large expected move already — the position needs the *actual* move to exceed what's already priced (roughly, the combined premium as a % of spot: $58/1240 \approx 4.7\%$), and is additionally exposed to an IV-crush loss (Part 5.9) if the actual move disappoints even directionally. - **Breakevens:** $1240 + 58 = ₹1,298$ (up) and $1240 - 58 = ₹1,182$ (down). - **Risk:** capped at the ₹58/share premium paid (known, bounded loss) if the stock closes between the breakevens. - **Position size:** sized so the *maximum* loss (full premium paid, a realistic outcome for a straddle on a quiet result) still respects the account's 1-2% per-trade risk rule from Part 8.3 — treating the entire premium as "at risk" capital, not assuming a partial, softer loss the way a stop-managed directional trade might. - **Exit plan:** exit immediately post-results on the opening move rather than holding into further time decay — once the event that justified the trade has occurred, theta (Part 5.4) becomes a pure headwind with no further edge-justifying catalyst ahead.

PART 17 — SEASONALITY, CALENDAR EFFECTS & MARKET CYCLES

17.1 Why seasonality is treated as a legitimate (if secondary) input, not superstition

Seasonality refers to statistically-recurring patterns in returns tied to the calendar — a specific month, day of week, or time-of-month — that persist across many years with enough consistency to be more than random noise, though generally with a smaller, less reliable edge than a clear trend or momentum signal. A disciplined TRA treats seasonality as one input among several (confluence, per Part 3.7's golden rule), never a standalone trading trigger.

17.2 Documented seasonal patterns relevant to Indian markets

- **The "Santa Claus rally" / calendar-year-end effect:** many markets, including Indian indices, show a modest tendency toward positive returns in the last days of December and first days of January, commonly attributed to institutional portfolio positioning, year-end fund flows, and reduced selling pressure during low-liquidity holiday periods.
- **Muhurat trading:** a symbolic one-hour trading session on Diwali evening, considered auspicious for initiating new positions in Indian markets — more a cultural/sentiment phenomenon than a statistically robust edge, but one every India-focused TRA should know by name given how frequently it comes up in year-end market commentary.
- **Result-season clustering:** Indian quarterly-earnings seasons (roughly the weeks following each quarter-end) see systematically elevated volatility and volume as the majority of large-cap results cluster into a compressed few weeks — a TRA should expect materially different average daily ranges and gap risk during result season versus a quiet mid-quarter month, and size intraday positions accordingly.
- **Monthly F&O expiry effects:** the days immediately around monthly (and, for major indices, weekly) derivatives expiry often show distinct volatility and pinning behaviour (Part 5.6's Max Pain discussion) as large option positions are unwound or rolled — a well-documented, tradeable microstructure pattern distinct from broader calendar seasonality.

17.3 Market cycles and sector rotation, extended

Building on Part 4.4's brief mention: different sectors historically lead and lag at different stages of the broader economic/market cycle — early-cycle recovery phases often favour rate-sensitive and cyclical sectors (banking, real estate, autos) as growth expectations improve from a low base; late-cycle phases often favour defensives (FMCG, pharma, utilities) as growth expectations peak and investors rotate toward earnings stability. A TRA tracking **relative strength** of sector indices against the broader market (Nifty Bank vs Nifty 50, for instance) over rolling periods can identify rotation *as it happens* in the price data, ahead of it being widely discussed in fundamental commentary — a genuinely technical-analysis-native way to read the macro cycle.

17.4 The honest limits of seasonality as an edge

Seasonal patterns are statistical tendencies estimated from a finite (and shrinking, as market structure evolves) historical sample, and — like the anomalies discussed in the fundamental-analysis literature — can weaken or disappear once widely known and traded on (the same "arbitraged away" logic from market-

efficiency theory applies here too). A TRA should present seasonality as a mild, confirming input to a thesis already supported by trend/level/momentum confluence, never as the primary justification for a trade — an interviewer asking "how much do you rely on seasonality" is testing exactly this calibration.

PART 18 — ALGORITHMIC EXECUTION & MARKET MICROSTRUCTURE

18.1 Why execution mechanics matter to a technical research analyst, not just a trader

A TRA's chart-based signal is only as good as the price actually achieved when acting on it — a signal correctly identified but poorly executed (large market impact, adverse slippage) can turn a good call into a losing trade even when the underlying analysis was right. Understanding how large orders are actually worked in the market is directly relevant to translating a research call into a real, executable recommendation, especially for less liquid names.

18.2 VWAP and TWAP — the two standard execution benchmarks

- **VWAP (Volume-Weighted Average Price):** the average price of a security over a period, weighted by volume traded at each price level — the standard benchmark institutional desks are measured against ("did we execute better or worse than VWAP"), since it reflects where the *bulk* of trading actually happened, not just a simple time-average.
- **TWAP (Time-Weighted Average Price):** the simple average price over evenly-spaced time intervals, regardless of volume — used when a desk wants to spread execution evenly across a time window irrespective of volume patterns, typically for less liquid names where volume-weighting could concentrate too much of the order into a short, thin window.
- **Algo execution strategies** built around these benchmarks (a "VWAP algo," a "TWAP algo") automatically slice a large order into smaller pieces released over time to minimise market impact and tracking error against the chosen benchmark — a TRA recommending a large position in a mid/small-cap name should flag the expected execution approach and realistic achievable price, not just the current quoted price, since a large order in a thin name can move the market meaningfully against the trader while filling.

18.3 Iceberg orders and hidden liquidity

An **iceberg order** displays only a small portion of a much larger total order size to the public order book, automatically replenishing the visible portion as it fills — used specifically to avoid signalling a large order's true size to the rest of the market, which could otherwise cause other participants to front-run or adjust their own pricing against the visible large order. A TRA analysing order-book depth (Part 2.2's support/resistance-from-order-flow discussion) should be aware that visible depth can meaningfully understate real institutional interest at a level, since iceberg orders deliberately hide the bulk of their size.

18.4 Market impact and why "the chart says buy" isn't the whole execution story

Continuing the market-impact concept from the broader capital-markets literature: a large order in a thin, illiquid name "walks the book," achieving a materially worse average price than the quoted best bid/ask suggests. A TRA's recommendation for a large or institutional-sized position in a less liquid name should explicitly account for this — either by recommending a longer execution window (accepting price-timing risk in exchange for lower market impact) or by sizing the recommended position down to what the name's actual liquidity can absorb without excessive impact, a genuinely practical skill that separates research recommendations useful to an actual trading desk from a purely theoretical chart call.

PART 19 — GLOBAL MARKET CUES & GAP TRADING

19.1 Why Indian markets open with a built-in overnight information gap

NSE/BSE cash-market trading hours (9:15am-3:30pm IST) sit in a window where major US markets have already closed for the day (US markets close around 1:30-2:30am IST depending on daylight saving) and European markets are only just opening — meaning every Indian trading session opens carrying a full overnight's worth of global information (US market close levels, overnight commodity/currency moves, any major news) that couldn't be traded on domestically until the next session opens. This structural gap is why "global cues" is a fixed, non-optional line item in every professional pre-market note (Part 9.1).

19.2 The standard pre-market global-cues checklist

- **US market close** (S&P 500, Nasdaq, Dow) — the single most-watched overnight reference, since US market direction has historically shown meaningful correlation with next-day Indian market sentiment, particularly for globally-linked sectors (IT services, given US client exposure).
- **Asian markets during Indian pre-market hours** (Nikkei, Hang Seng, and other Asian indices trading concurrently with or just before the Indian pre-open) — the most contemporaneous read available, since these markets are live and reacting in real time as the Indian session approaches.
- **SGX Nifty / GIFT Nifty**: a Nifty-linked contract trading on an international exchange during hours the Indian market itself is closed, historically used as the single best real-time proxy for where the Indian market will likely open — a TRA's pre-market note leans heavily on this specific data point.
- **Crude oil overnight move**: material for India specifically given its large net-oil-import position (Part 15.2's intermarket discussion) — an overnight crude spike is a standard, direct input to the pre-market view on rate-sensitive and inflation-exposed sectors.
- **US 10-year Treasury yield and Dollar Index (DXY) overnight moves**: feed directly into the intermarket framework from Part 15 — rising US yields or a strengthening dollar are typically read as headwinds for emerging-market equities including India, all else equal.
- **Any overnight company-specific news**: US-listed ADRs of Indian companies (where they exist) trading overnight can itself be an early, if imperfect, signal for that specific stock's likely opening move.

19.3 Gap trading — reading and trading the opening gap itself

A **gap** is the difference between the previous close and the current session's open, driven by the overnight information (Part 19.1-19.2) being priced in all at once at the open rather than continuously through the prior session. - **Gap types**: a **common gap** (small, within recent average range, often fills quickly and carries little signal) versus a **breakaway gap** (a large gap on a fresh trend-initiating catalyst, often *not* filled quickly and can mark the start of a sustained move) versus an **exhaustion gap** (a large gap late in an extended trend, often marking a climactic, unsustainable final move before a reversal) — distinguishing these three types by context (where the gap occurs relative to the existing trend and recent range) is a core, frequently-tested TRA skill. - **"Gap and go" vs "gap fill" as two distinct trading approaches**: a gap-and-go trader treats a strong breakaway gap as a signal to trade in the gap's direction, expecting continuation; a gap-fill trader treats a common or exhaustion gap as likely to partially or fully retrace back toward the prior close before the underlying trend (if any) reasserts — the two approaches are not contradictory, they apply to different gap *types*, and correctly classifying the gap (Part 19.3's three types) is what determines which approach actually applies to a given morning's gap.

19.4 Worked example — synthesising a full pre-market view

Overnight: S&P 500 closed +0.8%, crude oil fell 2.5%, US 10-year yield fell 8bps, GIFT Nifty is indicating a flat-to-slightly-positive open. A specific IT-services stock under coverage has no company-specific overnight news.

Model answer. Positive US close and falling yields are mildly supportive for Indian equities broadly (Part 15.2/15.3's intermarket logic); falling crude is supportive for the rupee and inflation-sensitive sectors specifically, though less directly relevant to an IT-services name; GIFT Nifty's flat-to-slightly-positive indication suggests the broad market itself isn't expected to gap meaningfully. For the specific IT-services stock with no company-specific news, the reasonable pre-market view is that it should open roughly in line with the broader index's expected flat-to-slightly-positive move, with the stock's own technical levels (Part 1-3) — not the overnight cues, which are broadly neutral-to-mildly-positive here — doing most of the work in determining the actual intraday trade plan. This is the disciplined synthesis a professional pre-market note performs every single morning: named inputs, explicit read on each, and a clear statement of how much (or how little) the overnight picture actually changes the existing technical view.

PART 20 — SECTOR & THEMATIC INDEX TRADING

20.1 Why sector indices deserve their own dedicated technical approach

Bank Nifty, Nifty IT, Nifty Auto, Nifty Pharma and other sectoral indices each have their own distinct volatility signature, dominant-stock concentration, and macro sensitivity — treating them as smaller versions of the broad Nifty 50 misses tradeable structural differences a specialised TRA should know cold, especially given how heavily traded Bank Nifty derivatives specifically are in the Indian market.

20.2 Bank Nifty — concentration, rate-sensitivity, and its own volatility character

Bank Nifty is dominated by a small number of large private and PSU banks, meaning single-stock news on the top 2-3 constituents can move the entire index disproportionately compared to the broader, more diversified Nifty 50 — a TRA trading Bank Nifty options must track index-heavyweight-specific news (a top private bank's results, an RBI policy surprise affecting the banking sector specifically) with the same diligence as broad-market cues. Bank Nifty also typically carries higher implied volatility than Nifty 50 (reflecting both its higher realised volatility and its concentration), meaning options strategies calibrated for Nifty's typical IV level often need adjustment — a straddle sized using Nifty's usual expected-move assumptions will be miscalibrated if applied unchanged to Bank Nifty.

20.3 Nifty IT — a genuinely different macro driver set than the domestic-facing sectors

Nifty IT's constituents derive the majority of their revenue from US/European clients, making the sector unusually sensitive to **USD/INR** (a weaker rupee is a direct earnings tailwind for dollar-revenue, rupee-cost businesses) and to **US technology-sector demand cycles and corporate IT-spending trends** — arguably more sensitive to US-specific intermarket cues (Part 15, Part 19.2) than to purely domestic Indian macro data, a genuinely distinct driver profile from Bank Nifty's domestic-rate-and-credit-cycle sensitivity. A TRA covering Nifty IT should weight US technology-sector earnings season and Fed policy commentary alongside the standard domestic pre-market checklist.

20.4 Thematic and strategy indices — momentum, quality, low-volatility factor indices

Beyond sector indices, India has increasingly liquid factor/thematic indices (Nifty200 Momentum, Nifty Quality, Nifty Low Volatility, and similar) that are constructed using explicit rules (e.g. selecting and weighting constituents by trailing price momentum) rather than simple market-cap weighting — these give a TRA a direct, tradeable way to express a factor view (e.g. "momentum is working in this market regime") without having to build a custom stock-picking screen, and their own historical performance patterns (which factors have led in which macro regimes) are a legitimate extension of the sector-rotation framework from Part 17.3.

20.5 Worked example — a relative-strength-based sector rotation call

Nifty Bank has outperformed Nifty IT by 8% over the trailing 3 months, with Nifty Bank making new relative highs against Nifty IT on the ratio chart ($\text{Nifty Bank} \div \text{Nifty IT}$), while Nifty IT's absolute chart shows a weakening trend below its 50-day MA.

Model answer. Plotting the ratio of two sector indices against each other (a "relative strength" or "ratio" chart) isolates which sector is outperforming independent of the broader market's overall direction — a rising Nifty Bank/Nifty IT ratio making new highs indicates money is rotating toward banking and away from IT specifically, consistent with Part 17.3's cycle-rotation logic if the macro backdrop (e.g. domestic credit growth accelerating, global tech-spending cooling per Part 20.3's driver set) supports that read. The combined signal — relative strength confirming the rotation, plus Nifty IT's own absolute technical weakness (below its 50-day MA) — is a stronger basis for a pairs-style rotation trade (long Bank Nifty, short or underweight Nifty IT) than either signal alone, exactly the cross-index confluence approach a sector-focused TRA is expected to apply.

PART 21 — VOLATILITY TRADING & INDIA VIX-BASED STRATEGIES

21.1 Volatility as a tradeable asset in its own right, not just an options-pricing input

Part 5.5 introduced implied volatility as an input to option pricing. This Part treats **volatility itself** as something a TRA can form a directional view on and trade — a genuinely distinct skill from having a view on price direction, since a trader can be right about volatility (it will rise or fall) while being agnostic or even wrong about which direction the underlying moves.

21.2 India VIX mechanics — what it actually measures

India VIX is calculated from the order book of Nifty index options (specifically, a weighted average of implied volatilities across a range of out-of-the-money Nifty options near-term and next-term expiries), expressing the market's expectation of Nifty's annualised volatility over the next 30 days. A VIX reading of 15 implies the market expects roughly a 15% annualised standard deviation of Nifty returns — converting to a rough expected 30-day move via $15\% \times \sqrt{30/365} \approx 4.3\%$, a quick mental-math conversion worth having memorised for a live interview question.

21.3 The VIX-Nifty inverse relationship and what breaks it

India VIX and Nifty typically move inversely — Nifty selloffs are usually accompanied by VIX spikes (fear/hedging demand pushes option premiums and thus implied volatility up as the index falls), while calm, grinding uptrends see VIX drift lower. This relationship is strong but not perfect: VIX can rise even during a rally if the *market structure* itself becomes more uncertain (heavy event risk ahead — an election, a major central-bank decision — can keep VIX elevated even while price grinds higher), and recognising this decoupling as a signal in itself (elevated VIX despite a rising market = the market is pricing meaningful event risk ahead, not complacency) is a more sophisticated read than assuming the inverse relationship always holds mechanically.

21.4 Trading the volatility view directly — long and short vega strategies

- **Long volatility** (buying straddles/strangles, Part 5.7): profits if realised volatility ends up higher than what was implied when the position was opened, regardless of direction — the standard pre-event (results, policy announcement) volatility trade already covered in Part 5.9's straddle example.
- **Short volatility** (selling straddles/strangles/iron condors): profits if realised volatility ends up lower than what was implied — collecting the premium as compensation for underwriting the risk that the market moves less than the option prices suggested. Short-volatility strategies have a favourable win rate in typical, calm market regimes but carry the asymmetric tail risk flagged in Part 5.7 and the worked example in Part 16.4 of the interview appendix — a single large, unexpected move can erase many periods' worth of collected premium.
- **Volatility mean-reversion**: VIX itself has a strong historical tendency to mean-revert — extreme spikes (a crisis-level VIX reading) have historically been followed by a decline back toward more typical levels, and extended periods of unusually low VIX have historically preceded a return to higher volatility. A TRA monitoring VIX Rank/Percentile (the same concept from Part 5.5, applied to VIX's own historical range) uses this to time long-vs-short-volatility positioning, not just individual option trades.

21.5 Worked example — reading a volatility surface/term-structure signal

India VIX (spot, ~30-day expectation) is at 22, while a VIX futures-style implied reading for 3 months out sits at 17 — a downward-sloping ("backwardated") volatility term structure.

Model answer. A backwardated volatility term structure — near-term implied volatility higher than longer-dated — typically signals the market is pricing a specific, near-term risk event or already-elevated stress that's expected to *subside* over the following months, rather than a persistent, structurally higher volatility regime (which would instead show a flatter or upward-sloping term structure). This is analogous to the futures contango/backwardation concept (Part 5.2) applied to volatility instead of price — and practically, it suggests a short-near-term/long-further-dated volatility calendar structure could be considered by a sophisticated volatility trader expecting the elevated near-term reading to normalise, though the specific near-term catalyst driving the elevated 22 reading should always be identified explicitly (an election, a major results date, a global risk event) rather than trading the term-structure shape in isolation without understanding what's actually driving it.

APPENDIX B — INTERVIEW Q&A (THEORY + WORKED)

1. **Q: What's the difference between technical, fundamental, and quantitative analysis, in one line each?** A: Fundamental answers *what* to buy (business value); technical answers *when* (price/volume timing); quantitative uses statistics/code to systematically find and validate an edge (often combining elements of both).
2. **Q: Why does Dow Theory require volume to confirm a trend?** A: Price movement on weak volume reflects a thin, low-conviction market that can reverse easily; volume expanding in the trend's direction shows broad participant conviction behind the move, making the trend more likely to persist — this is why a breakout on light volume (Part 2 chart-pattern content) is treated with suspicion.
3. **Q: A stock's RSI shows bearish divergence (price makes a new high, RSI makes a lower high) at a known resistance level. How would you construct the trade?** A: This is a confluence setup (trend/level + momentum divergence) suggesting weakening momentum right at a level where sellers have historically emerged — a short/reduce-exposure candidate with an entry on confirmation (e.g. a bearish reversal candle), a stop just above the resistance/recent high, and a target at the next support — always sized per the fixed-risk-% rule (Part 8.3), never on conviction alone.
4. **Q: Explain Max Pain and its practical use, including its limits.** A: Max Pain is the strike price at which the largest number of options (by value) would expire worthless, theoretically the point option writers "prefer" and price sometimes gravitates toward near expiry due to hedging flows from large option writers. Its limit: it's a probabilistic tendency observed empirically, not a guarantee, and can be overridden by strong fundamental/news-driven moves — it should inform, not solely determine, an expiry-week view.
5. **Q: Worked — what does a rising Put-Call Ratio (PCR) from 0.6 to 1.3 over a week suggest, and what's the contrarian read?** A: A PCR rising from 0.6 (call-heavy, bullish positioning) to 1.3 (put-heavy, bearish positioning) suggests growing bearish sentiment/hedging in the options market. The contrarian read many technical analysts apply: an extremely high PCR can signal excessive bearishness/hedging that's already priced in, sometimes preceding a relief rally, particularly if it coincides with price holding above a known support level rather than breaking it — PCR is best read alongside price action, not in isolation.
6. **Q: Why is "no look-ahead bias" one of the most common backtesting errors, and give a concrete example of how it creeps in accidentally.** A: Look-ahead bias means a backtest accidentally uses information not actually available at the moment a real trading decision would have been made — e.g. a strategy using a day's closing price to generate a signal that's then assumed to be actionable at that same day's open, when in reality the closing price isn't known until after the market closes; the fix is lagging every signal by at least one full period so the backtest only ever "sees" what would genuinely have been known at decision time.
7. **Q: Why might a strategy with a higher CAGR have a worse Sharpe ratio than a strategy with a lower CAGR (as in the Part 10.5 worked example)?** A: Sharpe measures return *per unit of volatility taken*, not raw return — a higher-CAGR strategy that achieves its return via a bumpier, more volatile equity curve can have a worse (lower) Sharpe than a steadier, lower-CAGR strategy, because the denominator (volatility) grew faster than the numerator (excess return). A professional evaluates both, since a smoother, lower-CAGR strategy may be preferable for capital that can't tolerate large drawdowns.
8. **Q: What specifically must a SEBI-registered Research Analyst disclose, and why does this matter beyond just "following the rules"?** A: Any personal holding or financial interest in a

security they're recommending, and any conflict of interest (e.g. the analyst's firm has an investment-banking relationship with the company). This matters because undisclosed conflicts undermine the very reason clients pay attention to research in the first place — credibility — and disclosure is what lets a client correctly weight the analyst's view knowing the incentives behind it.

9. **Q: Worked — an options seller has collected ₹40,000 in premium across a month writing weekly iron condors, but one adverse week produces a ₹65,000 loss. Was the strategy "wrong"?** A: Not necessarily — premium-selling/income strategies (like iron condors) typically have a high win rate but an asymmetric payoff (many small wins, occasional larger losses when price moves outside the range), which is the mirror image of a long-option buyer's payoff (many small losses, occasional large win). A single adverse week doesn't invalidate the strategy; what matters is whether the *sized* risk on each individual position respected account risk rules (Part 8) and whether the cumulative expectancy (Part 10.4) across a large enough sample is positive — the same discipline of judging a strategy by process and expectancy, not any single outcome, that Part 8's "survive first" principle emphasises.
10. **Q: What's the difference between IV Rank and IV Percentile, and why might a trader care about the distinction?** A: IV Rank measures where current IV sits between the highest and lowest IV over a lookback period (a simple range-based measure); IV Percentile measures the percentage of days in that lookback period where IV was *lower* than the current level (a distribution-based measure) — the two can genuinely differ (e.g. IV could be near the top of its range by Rank, but only in, say, the 60th Percentile if IV spent a lot of time clustered near that high level) — Percentile is generally considered the more robust measure since it isn't distorted by a single historical outlier high or low the way Rank can be.
11. **Q: How would you structure an answer to "walk me through your process for a single trade idea," combining multiple parts of this handbook?** A: State the structure explicitly: (1) higher-timeframe trend context (Part 1.6, 1.4), (2) key levels — support/resistance from price structure and, for F&O names, from OI (Part 5.6), (3) price-action confirmation at that level (candlestick/pattern, Part 2), (4) indicator confluence — one trend + one momentum + volume, checking for divergence (Part 3.7, 4.1), (5) for derivatives specifically, a check of IV/OI/PCR context (Part 5.5-5.6), (6) a fully defined trade — entry, stop, target, R:R, horizon, and (7) correctly sized per the fixed-risk-% rule (Part 8.3). Walking through this structure explicitly, rather than describing an ad hoc impression of a chart, is exactly what the "Anatomy of a good call" section (9.2) and this appendix's worked examples (A.1-A.2) are modelling.
12. **Q: Why does the handbook insist a call is unprofessional without a stated stop-loss, even for a high-conviction idea?** A: A stop-loss is the mechanism that bounds risk to a known, pre-decided amount *before* the trade is placed — without one, a losing trade has no defined exit discipline and can compound into a much larger loss than any reasonable position-sizing calculation assumed, silently invalidating the entire risk-management framework (Part 8) the position size itself was built on. High conviction is not a substitute for a defined worst-case exit; it should, if anything, only affect position size within the fixed-risk-% rule, never remove the stop entirely.

End of handbook. Read it twice; the second pass is where it clicks. Pair this with the one-night crash course ([INTERVIEW_PREP_STUDY_GUIDE.pdf](#)) for the rapid revision version.